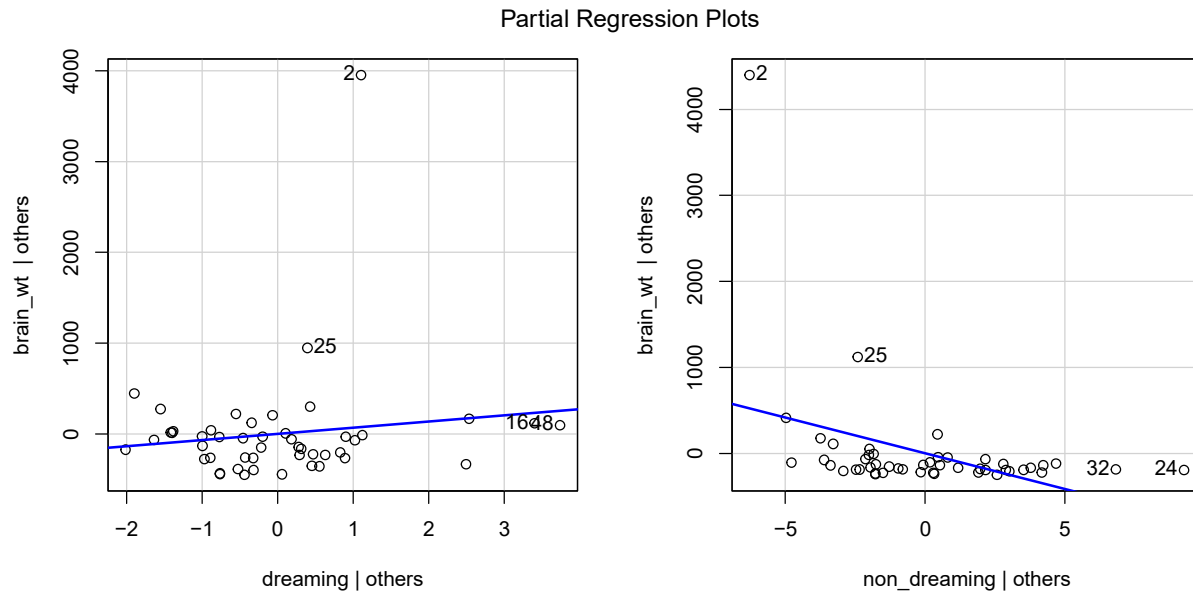


Q3.3 Linear Conditional Expectation

The relationship between the predictors and the outcome should be linear. We can check this using partial regression plots.



The partial regression plot for non_dreaming shows a reasonable positive linear trend. However, the plot for dreaming reveals a much weaker relationship, with the trend being heavily influenced by a number of outliers.

The assumption seems unlikely. The relationship with non_dreaming appears roughly linear, but the relationship with dreaming is weak and non-linear with it's litany of outliers.